

Day 7 – Reflection: LLM Tool Behaviors & Trust

Top 3 Surprises

1. Confidence ≠ Accuracy

The polished answer is often the most dangerous. All three tools—Perplexity, Claude, and ChatGPT—confidently produce wrong answers. The more complete and well-reasoned a response sounds, the more suspicious it should be. Readable, logical reasoning can be entirely wrong, inverting the intuition that clear thinking guarantees reliability.

2. Reasoning ≠ Accuracy

Claude demonstrates beautiful chains of logic while landing on false details. Reasoning transparency *feels* trustworthy but does not guarantee correctness. The ability to explain your work is distinct from the accuracy of the work itself. This is perhaps the most insidious failure mode because it's easiest to trust.

3. Citations Don't Mean Validation

Perplexity's source citations increase perceived reliability, yet the cited links often don't fully support the claims being made. Citation presence is a reliability signal, not proof. The combination of fast answers + attributed sources creates a credibility illusion that makes hallucinations harder to detect.

Top 3 Frustrations

1. Context Degradation Is Systemic

Long documents, extended conversations, and edge cases reliably break every model. None of them fail loudly—they simply drift into confident nonsense. This suggests the limitation isn't fixable by choosing a better tool; it's architectural.

2. Silent Outdated Knowledge

ChatGPT frequently presents outdated or incomplete information without flagging uncertainty. This is especially problematic for technical architecture and recent developments. The absence of a knowledge cutoff announcement means users can't know when they're working with stale information.

3. Shallow Synthesis Under Pressure

Perplexity collapses nuance rapidly. It mixes facts with assumptions and presents them as unified answers, which is fatal for complex or high-stakes reasoning. Speed comes at the cost of depth, but users may not realize the tradeoff.

Tool I Trust Least: Perplexity

Why: Not because it's useless, but because its failure mode is subtle and dangerous.

Perplexity blends real facts, speculative reasoning, and weakly relevant sources into a single authoritative answer. The combination of speed and citations creates a false sense of safety, making its hallucinations harder to detect than Claude's transparent over-caveating or ChatGPT's source-free certainty.

The danger: You get a fast, cited answer that *feels* verified but often isn't. For nuanced topics or anything you can't easily fact-check independently, this is particularly risky. The document explicitly warns against trusting Perplexity for complex reasoning, nuanced topics, and critical decisions.

Comparison:

- ChatGPT's confident-but-sourceless approach at least broadcasts its unreliability if you're paying attention
- Claude admits uncertainty upfront
- Perplexity does neither—it just cites sources that don't match the claims

Core Rule: No Tool Is Trustworthy Alone

- **Fast tools lie quickly**
- **Thoughtful tools lie slowly**
- **If the answer matters, assume the first response is wrong about something**
- **Force disagreement between tools**

The solution isn't finding the "best" tool. It's using multiple tools to verify each other, accepting that speed and accuracy are fundamentally opposed—not optimized—by any single model.

Decision Framework for Tool Selection

Need	Use	Why
Current facts	Perplexity (verify sources)	Speed + citations
Technical reasoning	Claude (verify critical claims)	Shows work, admits limits
Readable explanation	ChatGPT	Natural, clear writing
Unsure	Use two tools; disagreement = dig deeper	Cross-verification required

Key Takeaway

Confidence is the most dangerous failure mode across all tools. The polished, complete answer with sources is not safer—it's more dangerous because it's harder to question. Assume every first answer contains at least one significant error. Verification through multiple tools isn't optional for critical decisions; it's essenti